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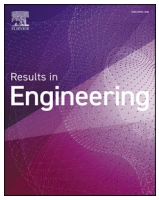

Comprehensive analysis of the properties of stainless steel 308l tungsten inert gas wire arc additive manufactured

Rauuf Nur Fattah [a], Iwan Istanto [a], Bambang Kusharjanta [a], Hammar Ilham Akbar [a], Yupiter H.P. Manurung [b], Moch Chamim [c], Triyono [a,*]

[a] Mechanical Engineering Department, Universitas Sebelas Maret, Surakarta, Indonesia
[b] Mechanical Engineering Department, Universiti Teknologi MARA, Shah Alam, Malaysia
[c] Mechanical Engineering Department, Sekolah Tinggi Teknologi "Warga", Surakarta, Indonesia

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ABSTRACT

This study provides a comprehensive evaluation of the microstructural evolution, mechanical properties, and corrosion behavior of 308 L stainless steel fabricated using Tungsten Inert Gas Wire Arc Additive Manufacturing (TIG-WAAM). Deposition was conducted using ER308L wire (1.0 mm diameter) with a CNC-controlled TIG-WAAM system, yielding a wall structure measuring 90 mm in height, 7 mm in thickness, and 125 mm in length. Although layer-by-layer deposition was consistently achieved, heat accumulation during the process caused dimensional distortion and collapse zones in the upper layers. Metallographic examination revealed a vertical grain evolution, from fine equiaxed grains at the bottom to coarse columnar grains at the top, accompanied by increased δ-ferrite content. This gradient correlated with surface hardness, which decreased from 242.5 HV (bottom) to 176.5 HV (top). Tensile tests indicated anisotropic behavior, with longitudinal specimens showing higher tensile strength (UTS ≈ 547 MPa) and elongation (≈ 35.5 %) than transverse specimens. Corrosion performance, assessed via Tafel polarization in 3.5 % NaCl solution, showed a distinct gradient in corrosion rate, from 0.0048 mm/year at the bottom to 0.1112 mm/year at the top, highlighting the influence of microstructural variation on electrochemical stability. Compared to previous studies on GMAW- and LP-based AM processes, this research uniquely incorporates a detailed corrosion analysis, addressing a critical gap in the literature. The integrated findings enhance understanding of TIG-WAAM 308L's performance, supporting its application in structural and corrosion-sensitive environments.

1. Introduction

Stainless steel 308 L is widely used across various industries, including automotive, medical devices, and general manufacturing, due to its excellent corrosion resistance and good weldability, which stem from its low carbon content and high chromium concentration [1]. Despite its favorable fabrication properties, forming SS 308 L often requires specialized equipment, as it is generally harder and tougher to machine compared to conventional carbon steels [2]. To overcome these challenges, Wire Arc Additive Manufacturing (WAAM) has emerged as a promising technique for processing stainless steel 308 L components with greater design flexibility [3–5].

WAAM is an additive manufacturing method based on electric arc welding, utilizing metal wire as feedstock to build structures layer by layer. Among various arc-based techniques, Gas Tungsten Arc Welding (GTAW), also known as Tungsten Inert Gas (TIG) welding, offers advantages in terms of arc stability and control over heat input [6,7]. Compared to traditional subtractive manufacturing methods, WAAM provides significant benefits, including higher material utilization, the ability to fabricate large-scale parts, and reduced production costs [8–11]. However, the high thermal energy involved in WAAM can lead to changes in microstructure and mechanical properties, particularly in components with complex geometries or large volumes. Consequently, optimizing WAAM process parameters has become a central focus of ongoing research efforts. Recent studies have employed advanced predictive and optimization approaches, such as bio-inspired neural networks and multi-objective optimization algorithms, to model and enhance process outcomes in WAAM of stainless steels [12,13]. These works highlight the growing trend toward integrating computational intelligence with experimental studies, reinforcing the importance of

* Corresponding author.
E-mail address: triyono74@staff.uns.ac.id (Triyono).

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systematic optimization in ensuring reliable microstructural and mechanical performance.

WAAM process parameters significantly influence bead shape, melt pool depth, and surface roughness, while shielding gas flow rate shows minimal effect on these characteristics [14]. For example, studies on Functionally Graded Materials (FGM) produced via WAAM using AISI 304 L and Alloy 82 reveal that the 304 L side exhibits lower hardness (184 ± 2 HV), ultimate tensile strength (565 ± 5 MPa), and yield strength (321 ± 3 MPa), with an elongation of approximately 23 ± 1.5 %. These structures also demonstrated relatively low impact toughness (33 ± 4 J) due to stress accumulation during each deposition cycle [15]. Furthermore, electrochemical corrosion testing and microstructural studies have shown that heat input and remelting during WAAM lead to anisotropic corrosion behavior, especially in SS304L [16].

While several studies have investigated the influence of WAAM parameters on microstructure, mechanical properties [17], tribological behavior [18], and corrosion resistance [19] in stainless steels, research specifically focused on TIG-WAAM of stainless steel 308 L remains limited. Most existing literature tends to explore AISI 304 L or other stainless alloys, leaving a gap in the understanding of how TIG-WAAM affects the integrated performance of 308 L. Moreover, a comprehensive evaluation that concurrently examines the mechanical properties, microstructure evolution, and corrosion resistance of TIG-WAAM SS 308 L is still lacking. Addressing this gap, the present study aims to provide a systematic and detailed analysis of SS 308 L fabricated using the TIG-WAAM process, highlighting the relationships between process parameters and resulting material properties. Specifically, the objectives of this work are threefold: (i) to investigate the influence of TIG-WAAM process parameters on the microstructural evolution of SS 308 L; (ii) to evaluate the resulting mechanical properties, including hardness, tensile strength, and impact toughness; and (iii) to assess the corrosion resistance behavior of TIG-WAAM fabricated SS 308 L. The findings of this work are expected to contribute to a deeper understanding of the suitability of this method for industrial applications and guide future process optimization strategies.

2. Methode

2.1. Materials

In this study, AISI 304 stainless steel plates with dimensions of 125 mm × 25 mm × 10 mm were used as the substrate. The filler material was ER308L wire with a diameter of 1.0 mm, manufactured by KISWEL Ltd. ER308L wire is commonly employed for welding austenitic stainless steels such as 304 L and similar alloys. This filler material was selected due to its excellent compatibility with the base metal, ensuring comparable corrosion resistance and mechanical performance [20].

The chemical compositions of the base metal and filler wire are provided in Table 1. The TIG-WAAM process was conducted using a layer-by-layer deposition approach, as illustrated in Fig. 1. Once the final geometry was achieved, the specimen was cut using a hacksaw for metallographic preparation, microhardness testing, and corrosion evaluation, while tensile test specimens were extracted using wire-cut EDM.

2.2. TIG-WAAM process and parameters

The specimen fabrication was performed using TIG-WAAM with a CNC-controlled system. An AOTAI inverter welding machine was employed to provide precise control of the torch movement along the X (horizontal) and Z (vertical) axes. The ER308L filler wire was fed continuously during deposition to build up the part layer-by-layer to the final dimensions of 90 mm in height and 125 mm in length.

The welding parameters in Table 2 were selected based on preliminary trials and previously reported studies on TIG-WAAM of stainless steels. The current (100 A) and voltage (15 V) provided sufficient heat input to achieve stable arc conditions and complete fusion of the filler wire with the substrate, while the welding speed (120 mm/min) and wire feed speed (250 mm/min) maintained consistent bead geometry and uniform deposition. The arc gap of 3 mm was chosen to ensure a stable arc column without excessive heat input, and an interpass time of 1 min was applied to control thermal accumulation between layers and reduce residual stress buildup. After deposition, the samples were left to cool naturally in ambient conditions to avoid thermal shock or residual stress buildup. Following solidification, the specimens were sectioned for metallographic examination, microhardness testing, corrosion testing, and tensile testing. Visual inspection was also conducted after each layer to monitor bead consistency and detect surface anomalies.

2.3. Microstructural and Microhardness analysis

Metallographic specimen preparation followed standard procedures, beginning with grinding using silicon carbide abrasive papers ranging from 100 to 2000 grit, followed by polishing with alumina paste and velvet cloth. The samples were etched using a solution of 10 ml dilute HNO_3, 10 ml HCl, and 10 ml distilled water for 10 s to reveal the microstructure. Microstructural observations were performed under 200 × magnification using optical microscopy.

Microhardness testing was conducted using the Vickers method with a 9.8 N load applied for 15 s. Measurements were taken at multiple locations across the specimen to evaluate hardness distribution throughout the deposited structure. Microhardness measurements were performed at three different cross-sectional regions of the TIG-WAAM wall, with at least five indents per region. Reported values represent the mean, and standard deviations were calculated to reflect data variability.

2.4. Corrosion rate testing (Three-Electrode method)

Corrosion behavior was evaluated using a three-electrode electrochemical cell. The TIG-WAAM specimen served as the working electrode, with an Ag/AgCl reference electrode and a platinum counter electrode. The test was conducted in a 3.5 % NaCl solution to simulate a marine corrosive environment, and a potentiostat was used to monitor and control voltage and current during the test. Each corrosion test was repeated three times under identical conditions, and the results were averaged.

Fig. 2 illustrates the schematic of the corrosion test setup and the area of observation. Prior to testing, the TIG-WAAM samples which have dimension of 15 mm x 15 mm x 5 mm were polished and cleaned to ensure surface consistency. Electrochemical measurements were carried out using either Tafel polarization or Linear Polarization Resistance (LPR) techniques to determine corrosion rates.

2.5. Tensile testing

Tensile test specimens were extracted from the TIG-WAAM-built wall with two different orientations: longitudinal (L) and transversal (T), as

Table 1
Chemical composition of materials [6].

Material	Fe	Cr	Ni	Mn	Si	C	P	S	Mo	V	Ti
AISI 304	Bal.	17.90	8.38	1.88	0.254	0.043	0.046	0.016	0.224	0.072	0.006
ER308L	Bal.	19.77	9.61	1.62	0.490	0.020	–	–	0.160	–	–

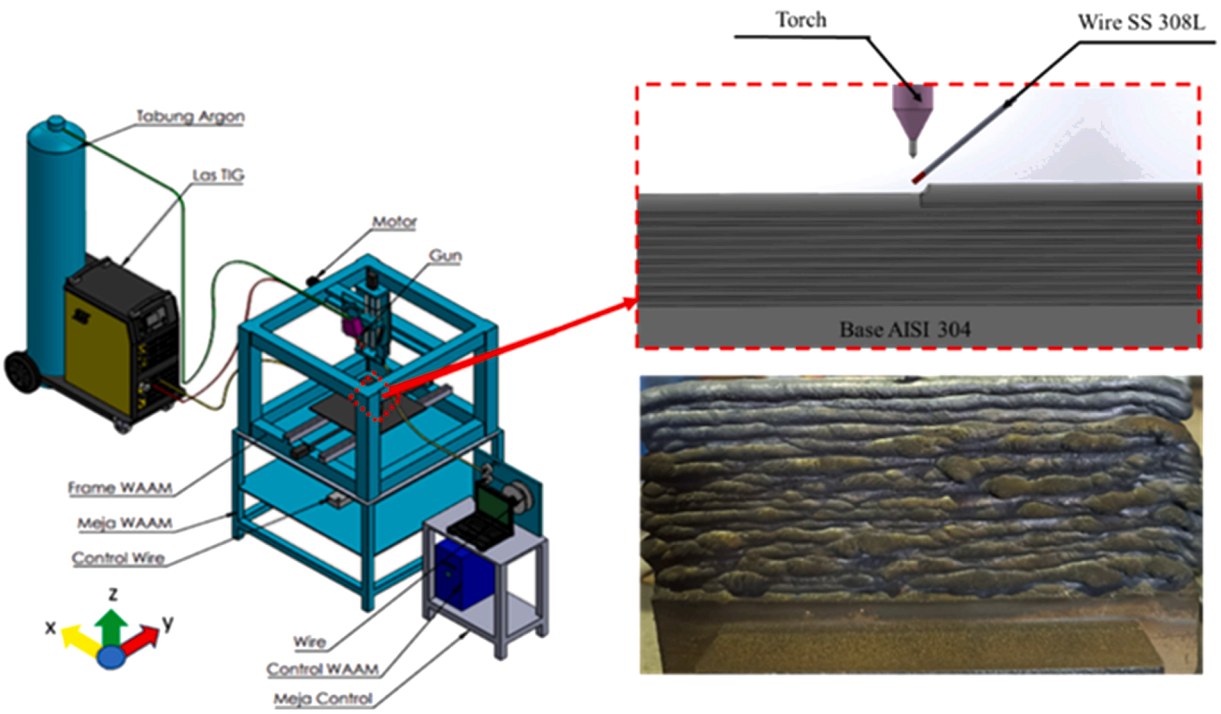


Fig. 1. Schematic of the TIG-WAAM.

Table 2
Welding process parameters.

Parameter	Value
Voltage	15 V
Current	100 A
Arc Gap	3 mm
Welding Speed	120 mm/min
Wire Feed Speed	250 mm/min
Interpass Time	1 min
Tungsten Electrode	2 % Ceriated, Ø 2.4 mm
Shielding Gas	Argon
Gas Flow Rate	15 L/min

shown in Fig. 3. This was done to evaluate the influence of deposition direction on the mechanical behavior. All specimens conformed to ASTM E8 standards. For each orientation, three or five tensile specimens were prepared and tested to ensure reproducibility. The specimens were clamped using precision jigs to ensure stability during testing. An axial tensile load was applied using a universal testing machine until fracture. Load and elongation data were recorded to generate stress–strain curves for each specimen.

3. Results and discussion

3.1. TIG-WAAM results

The Wire Arc Additive Manufacturing (WAAM) process using ER308L stainless steel wire via Gas Tungsten Arc Welding (GTAW) successfully produced a multilayered structure with consistent vertical deposition, as shown in Fig. 4(a). The printed structure exhibited clearly stacked weld beads, indicating that the wire feed rate and arc control were generally effective. The well-formed beads in the initial layers suggest adequate arc stability and controlled heat input during the early stages of deposition. However, closer inspection revealed a collapse zone in the middle to lower section of the structure, where bead uniformity deteriorated and excessive lateral metal spreading occurred. This deformation is symptomatic of interlayer heat accumulation, which can lead to partial remelting or over-softening of the underlying layers during the deposition of subsequent layers. Such thermal issues are common in WAAM processes, especially when heat dissipation is not properly managed between layers [21].

The after-grinded view in Fig. 4(b) shows a well-bonded interface between the deposited ER308L material and the AISI 304 substrate. The absence of macro defects such as large porosity or visible cracking indicates that the process provided adequate metallurgical bonding, at least on a macroscopic scale. This is consistent with reports showing that TIG-WAAM processes can produce defect-free interfaces when parameters are properly tuned [22]. The filler material adhered uniformly to the substrate, demonstrating good wettability and arc control in several regions. Nevertheless, the presence of localized collapse zones and distortion highlights the importance of thermal management strategies, such as the implementation of interpass cooling, optimized torch path planning, or real-time temperature monitoring, to enhance dimensional stability and overall structural integrity in WAAM builds [23,24].

Side views of the printed component (Fig. 4(c)) further revealed distortion effects, such as bending and flaring at the upper edges of the structure. These distortions arise due to thermal stresses generated by repeated heating and uneven cooling rates during the deposition process. The accumulation of residual stress and the lack of active cooling contribute to shape deviations from the intended geometry. Similar geometric inconsistencies and distortions have been reported in previous studies on WAAM of stainless steel and other high-temperature alloys, particularly in the absence of interpass temperature control or active cooling [25,26].

The variation in layer thickness also reflects instability in energy input and cooling behavior. Uneven bead heights suggest that process parameters such as travel speed, current, or wire feed rate might have fluctuated slightly during deposition. Maintaining consistent interpass temperature and optimizing dwell time are critical for reducing such variability, as emphasized in prior research on TIG-WAAM [27,28].

3.2. Metallographic analysis and surface hardness

In this study, a detailed metallographic analysis and microhardness

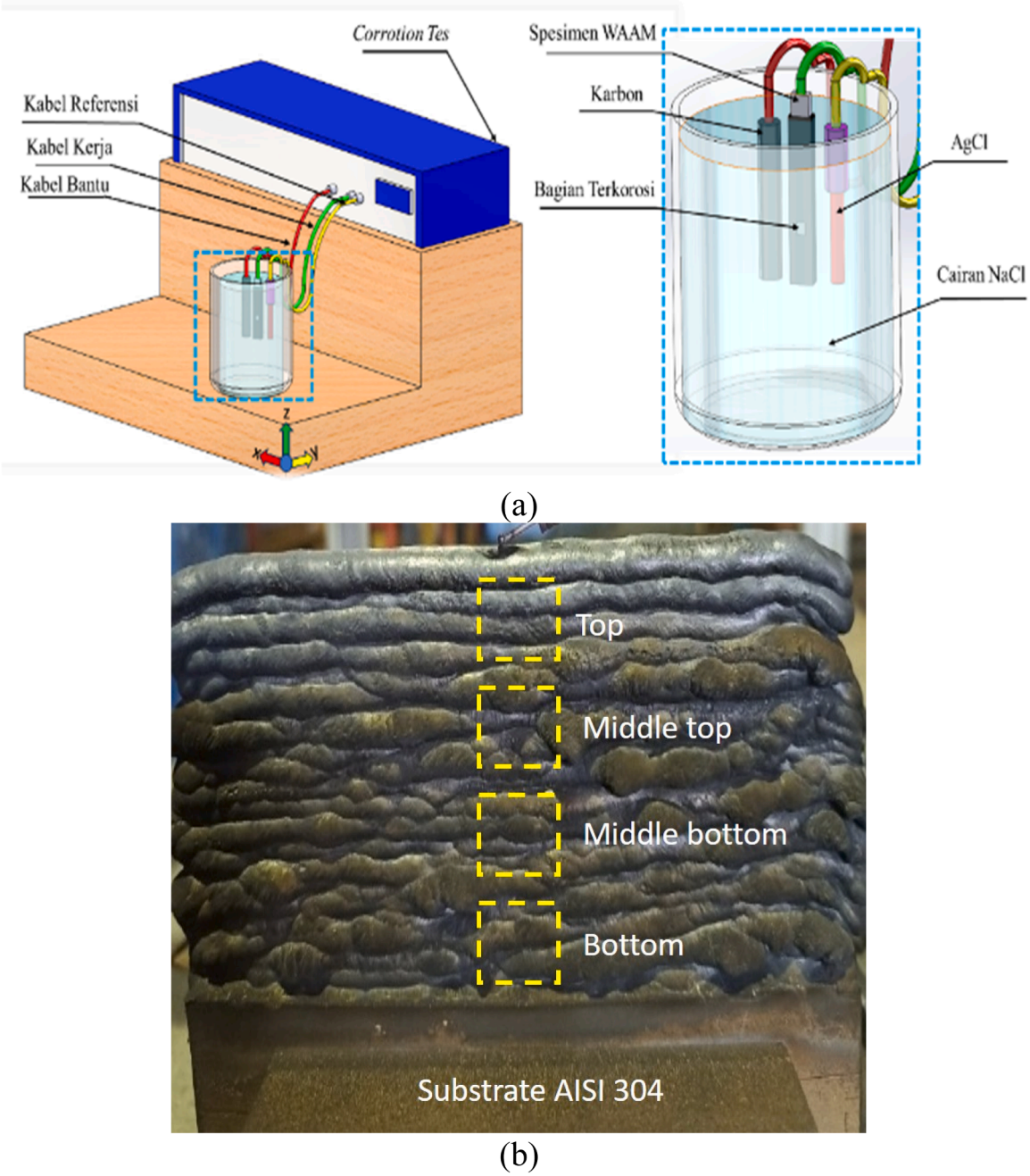


Fig. 2. The three-electrode corrosion testing setup, (a) schematic set up, (b) area of observation.

evaluation were performed on the WAAM-fabricated ER308L stainless steel specimens. The analysis was conducted across three regions of the printed wall: bottom, middle, and top, representing different thermal histories due to layer-by-layer depositionas seen in Fig. 5. Fig. 5(b) presents the macrostructure, which appears relatively uniform across the build, indicating consistent layer deposition. However, microscopic evaluation revealed distinct differences in grain morphology and hardness values along the build height, which can be attributed to variations in cooling rates. In the bottom region (Fig. 5(a) bottom), the microstructure exhibits fine dendritic patterns with a clear distribution of δ-ferrite within an austenitic (γ) matrix. This region, being closest to the heat sink (substrate), experienced the fastest cooling rate, leading to finer columnar grains and a higher degree of microstructural refinement. Consequently, this area recorded the highest hardness value of approximately 240 HV. Such microstructural refinement and enhanced hardness near the substrate are consistent with findings from previous WAAM studies on stainless steels [22–24].

The middle region (Fig. 5(a) middle) revealed coarser dendritic structures with larger grain sizes compared to the bottom. This zone experienced moderate thermal accumulation, resulting in slower cooling and allowing for greater grain growth. The hardness in this region dropped to around 180 HV, reflecting the typical inverse relationship between grain size and hardness described by the Hall–Petch effect [17]. The top region (Fig. 5(a) top) demonstrated the coarsest dendritic morphology, characterized by more equiaxed grains and reduced dendrite arm spacing. As the final layers receive continuous heat input without a significant heat sink above them, the cooling rate is the slowest in this region. This leads to further grain coarsening and a lower surface hardness of around 170 HV. Similar trends in grain coarsening and hardness reduction with build height have been reported in multiple

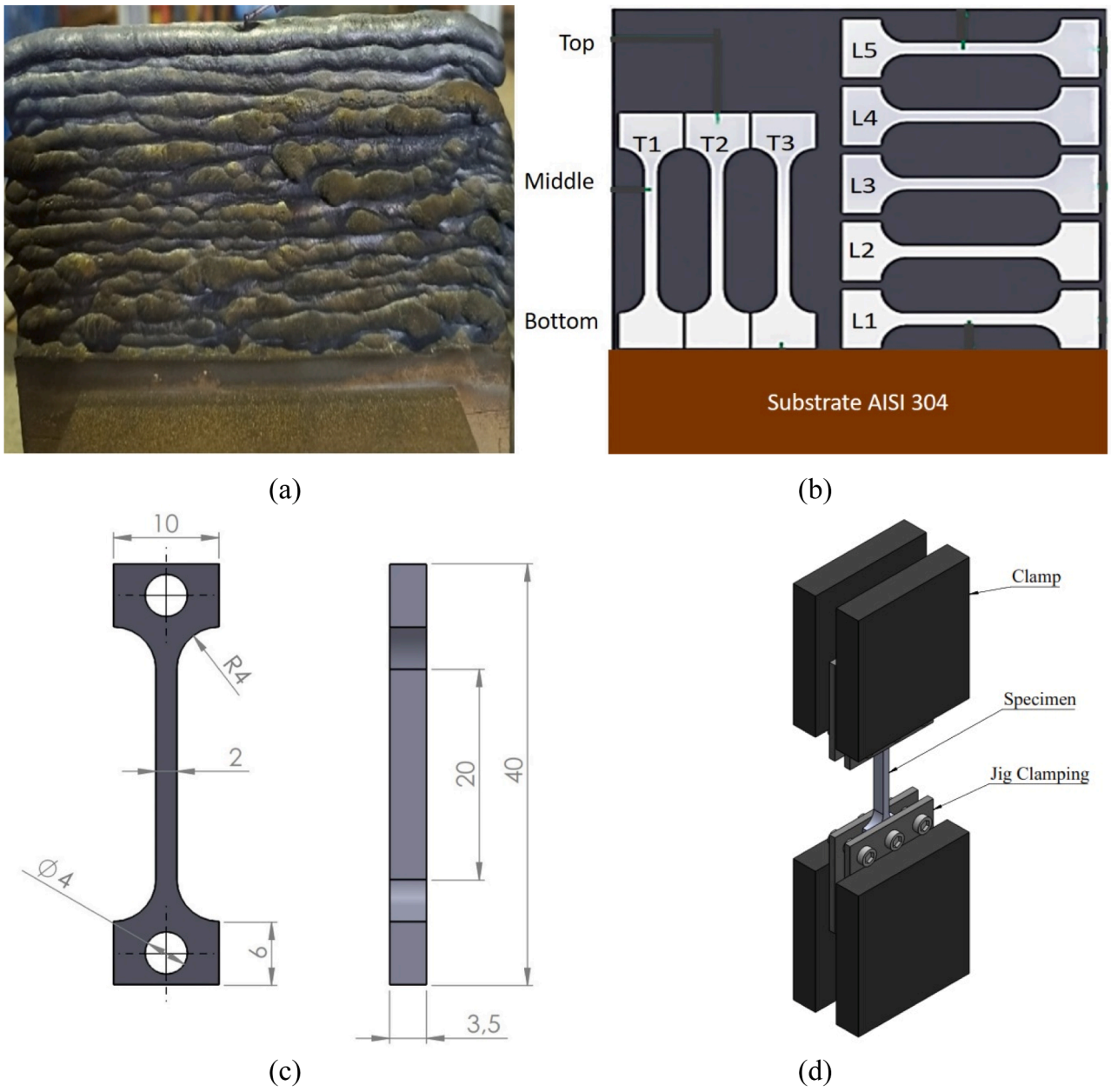


Fig. 3. Tensile test (a) WAAM building, (b) cutting position, (c) dimensions of tensile specimen (d) Schematic of tensile testing jig.

studies on austenitic stainless steels fabricated using WAAM and other arc-based additive manufacturing processes [16,22].

The correlation between microstructural morphology and hardness is clearly illustrated in Fig. 5(c), where the surface hardness decreases gradually from bottom to top. This trend supports prior research that emphasizes the role of thermal gradients and heat accumulation during WAAM in controlling grain structure and mechanical response [25]. Studies by Chamim et al. [6] and Sanjaya et al. [22] also report similar findings in TIG-WAAM-processed stainless steels, where thermal management significantly influences dendrite formation, grain boundary evolution, and local mechanical properties.

Moreover, literature confirms that faster cooling rates result in finer microstructures with superior hardness, whereas slower cooling promotes coarser grains and reduces hardness due to less effective suppression of grain growth [23,29]. Therefore, controlling interpass temperature, optimizing deposition speed, and applying localized cooling strategies could be critical to achieving uniform mechanical properties throughout WAAM-fabricated components. The observed variation in grain morphology and hardness along the build height is expected to have direct consequences on the functional properties of the WAAM component. Finer microstructures with higher hardness, such as those near the substrate, generally enhance both corrosion resistance and tensile strength, while coarser grains at the top may reduce these properties.

3.3. Corrosion test

Corrosion resistance is a key property of austenitic stainless steels, particularly for components intended for structural or marine environments. In this study, the corrosion behavior of TIG-WAAM-fabricated ER308L stainless steel was assessed using a three-electrode electrochemical cell in a 3.5 % NaCl solution, simulating a chloride-rich environment. The electrochemical response was analyzed using Tafel polarization curves, which provided the corrosion potential (Ecorr) and corrosion current density (Icorr) values, two essential parameters in evaluating corrosion performance. The results are summarized in Fig. 6.

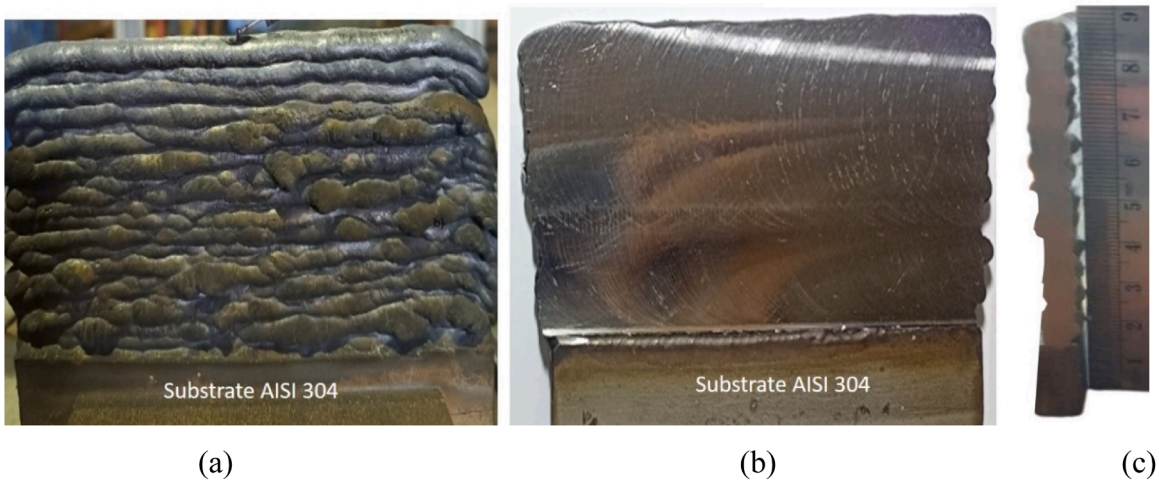


Fig. 4. (a) result of TIG-WAAM (b) after-grinded (c) distortion.

Since corrosion resistance in stainless steels is strongly influenced by microstructural features such as grain size, dendrite morphology, and phase balance, the previously observed hardness gradient provides important context. The superior hardness and refined microstructure at the bottom region contribute to better passivation and lower corrosion rates, whereas coarser grains and lower hardness at the top are expected to reduce corrosion resistance.

Tafel extrapolation revealed significant variation in corrosion behavior depending on the vertical position within the build (Fig. 6(a)). Based on the Fig. 6(b), the bottom section exhibited the lowest corrosion rate, approximately 0.0048 mm/year, indicating superior corrosion resistance. In contrast, the top section showed the highest corrosion rate, reaching 0.1112 mm/year. This indicates that the top layers are more susceptible to corrosion, likely due to microstructural coarsening, elemental segregation, or residual stress accumulation caused by repeated thermal cycling during deposition.

These findings align with the general understanding that corrosion resistance in stainless steels is closely tied to microstructural features, such as grain size, phase distribution, and ferrite content. According to Zhang et al. (2019), fine, homogeneously distributed dendritic microstructures, particularly with a balanced ferrite-austenite phase, enhance pitting resistance in chloride environments [29]. The bottom region of the WAAM build, having experienced rapid cooling, formed such refined structures, which likely contributed to its improved corrosion performance. In contrast, the top region, subjected to slower cooling and thermal accumulation, tends to develop coarser grains and potentially higher δ-ferrite content, both of which can adversely affect passive film stability [30]. Moreover, residual stresses, which are typically more pronounced in the top layers due to insufficient thermal dissipation, can also promote localized corrosion by disrupting the passive film and facilitating pitting initiation [31,32].

The results observed in this study are also in line with previous works by Behbahani et al. (2024), who reported that the corrosion resistance of WAAM-fabricated stainless steel declines along the build height due to changes in solidification dynamics and thermally induced segregation of alloying elements such as chromium and molybdenum [33]. Chromium depletion at grain boundaries or δ-ferrite phase boundaries may compromise the formation of a stable Cr2O3 passive layer, thus increasing vulnerability to corrosion [34–36]. These results underscore the necessity for thermal management and post-processing treatments, such as solution annealing or surface passivation, to homogenize the microstructure and restore corrosion resistance, especially in critical applications like marine structures, pipelines, or biomedical implants. Furthermore, the trends observed emphasize the need for site-specific property evaluations in WAAM-produced components. Assuming homogeneous properties across a build could lead to underestimating failure risks in service environments. Future work should incorporate post-weld heat treatments and electrochemical impedance spectroscopy (EIS) to provide deeper insight into the long-term corrosion behavior and passive film characteristics of WAAM 308 L stainless steel.

3.4. Tensile test

Tensile testing was conducted to assess the mechanical performance of ER308L stainless steel produced via the Wire Arc Additive Manufacturing (WAAM) process. Eight dogbone-shaped specimens were prepared with different build orientations: five longitudinal specimens (L1–L5), aligned parallel to the substrate, and three transverse specimens (T1–T3), oriented perpendicular to the substrate. The longitudinal specimens were further categorized by their vertical build position: bottom (L1, L2), middle (L3), and top (L4, L5).

The anisotropic mechanical behavior observed in WAAM builds is closely related to the microstructural gradients described earlier. Regions with finer grains and higher hardness typically exhibit greater strength and ductility, while coarser-grained areas are more prone to reduced tensile performance. Therefore, the tensile test results are discussed here in the context of the microstructure–hardness relationships and their implications for load-bearing applications

The stress–strain curves for all specimens are presented in Fig. 7, and the calculated values of ultimate tensile strength (UTS), yield strength (YS), and elongation are summarized in Fig. 8. These results reveal significant variations in mechanical properties depending on both the orientation and vertical position within the build. The highest UTS was recorded for the transverse specimen T3 at 533.43 MPa, while the longitudinal specimen L3 (mid-height) exhibited the highest UTS overall at 542.6 MPa. Longitudinal specimens from the top (L4, L5) and bottom (L1, L2) regions showed slightly lower average UTS values of 528.85 MPa and 527.07 MPa, respectively.

These variations are primarily attributed to the inherent anisotropy of the WAAM process, where mechanical properties depend strongly on build direction and thermal history. Along the build height, differences in cooling rates, from faster cooling near the substrate to slower cooling at the top, induce microstructural heterogeneity. Rapid cooling near the base promotes finer grains, while slower cooling at the top allows for grain coarsening. The middle region, benefitting from more balanced thermal cycles, typically yields a refined microstructure with improved mechanical performance.

As illustrated in Fig. 9, a pronounced distinction was observed between longitudinal and transverse specimens. Longitudinal specimens demonstrated superior mechanical properties, with an average YS of

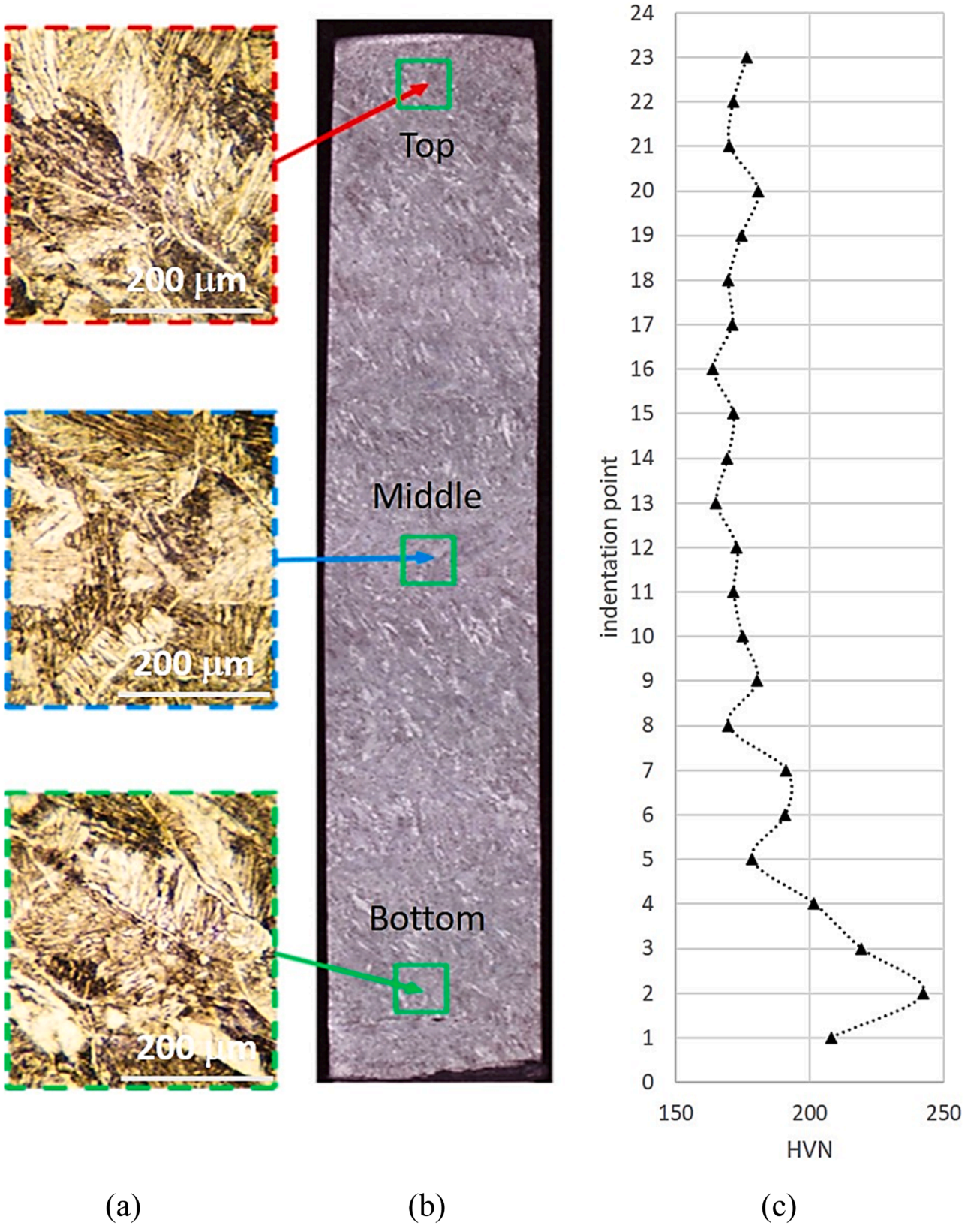
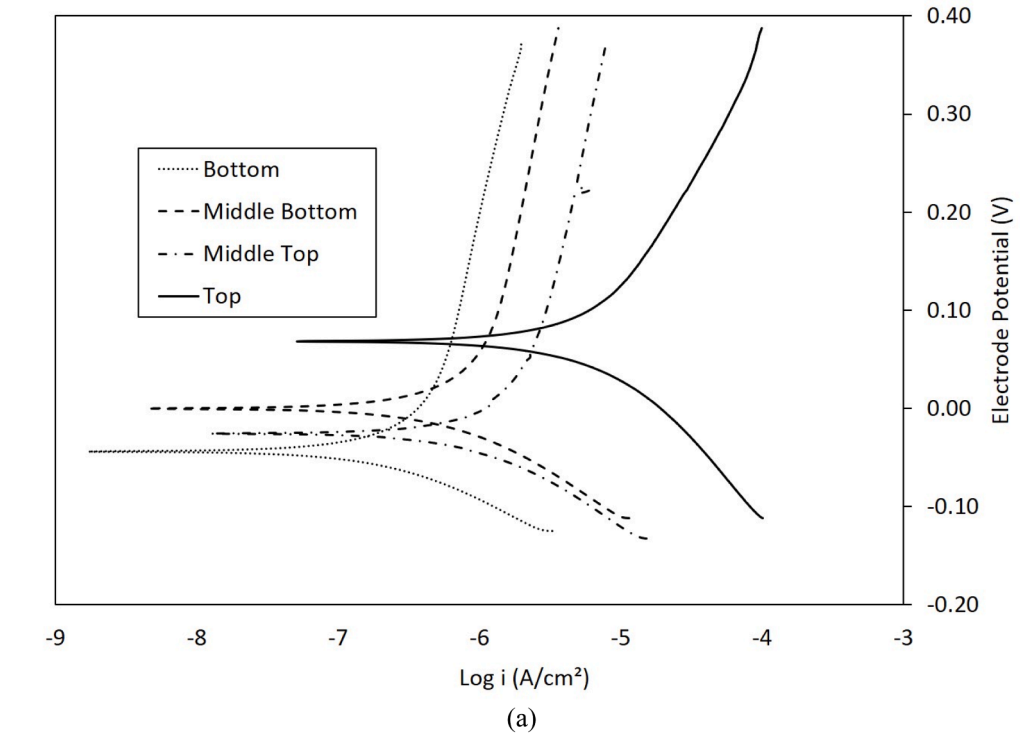


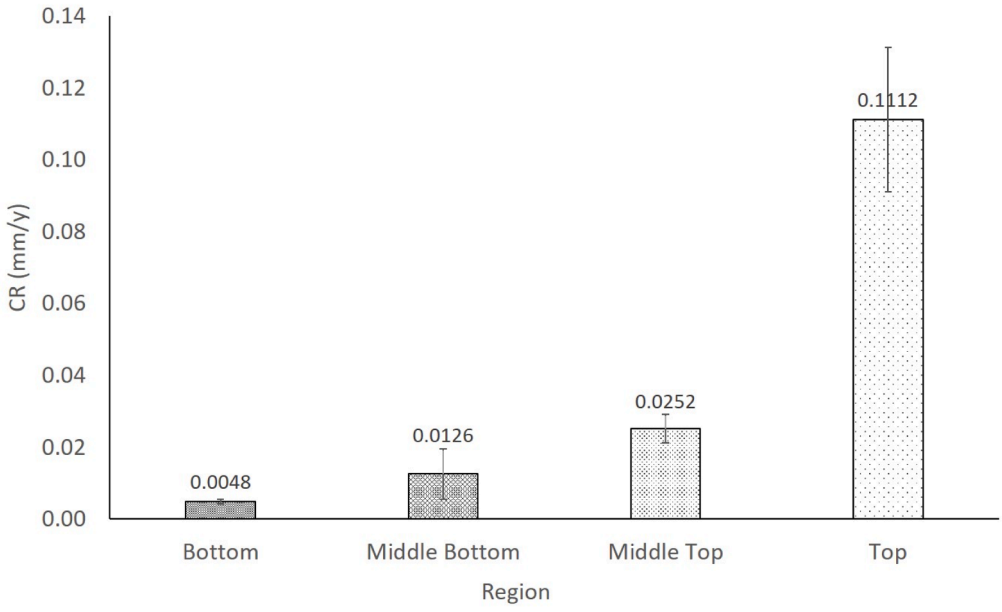
Fig. 5. (a) micro structure of bottom, middle and top region (b) macro structure, (e) hardness distribution.

364.4 MPa, UTS of 530.89 MPa, and elongation of 35.5 %. In contrast, transverse specimens exhibited lower performance, with YS of 306 MPa, UTS of 498.86 MPa, and elongation of only 19.1 %. This directional dependence aligns with prior studies. For example, Chen et al. (2024) reported that WAAM-fabricated stainless steel components often show enhanced strength and ductility along the build direction, due to elongated columnar grains aligned with the tensile load axis [37].

Fig. 10 illustrates macro-fracture morphologies. Longitudinal



(a)



(b)

Fig. 6. Corrosion behavior: (a) Tafel diagram, (b) corrosion rate.

specimens showed ductile fracture characteristics, with rough surfaces populated by numerous dimples, indicative of substantial plastic deformation before failure. This is typical for austenitic stainless steels, where ductility is favored by fine, equiaxed grains and lower residual stress levels. Conversely, transverse specimens exhibited smoother fracture surfaces with signs of brittle or quasi-brittle failure, suggesting limited ductile deformation. These differences emphasize the influence of build orientation and layer-by-layer deposition on the final mechanical integrity of WAAM components.

The presence of residual stresses, common in additive manufacturing, could also explain the reduced ductility in transverse directions. According to Wu et al. (2020), residual tensile stresses develop perpendicular to the deposition direction and can contribute to premature fracture or localized yielding under tensile loading [38]. Therefore, post-processing techniques such as stress-relief annealing or hot isostatic pressing (HIP) are often recommended to mitigate anisotropy and improve isotropic mechanical performance [39]. The tensile behavior of WAAM-fabricated SS308L exhibits strong orientation dependence, with superior strength and ductility in the longitudinal direction. These findings highlight the importance of considering build orientation in component design and the potential need for post-processing to ensure consistent mechanical performance in

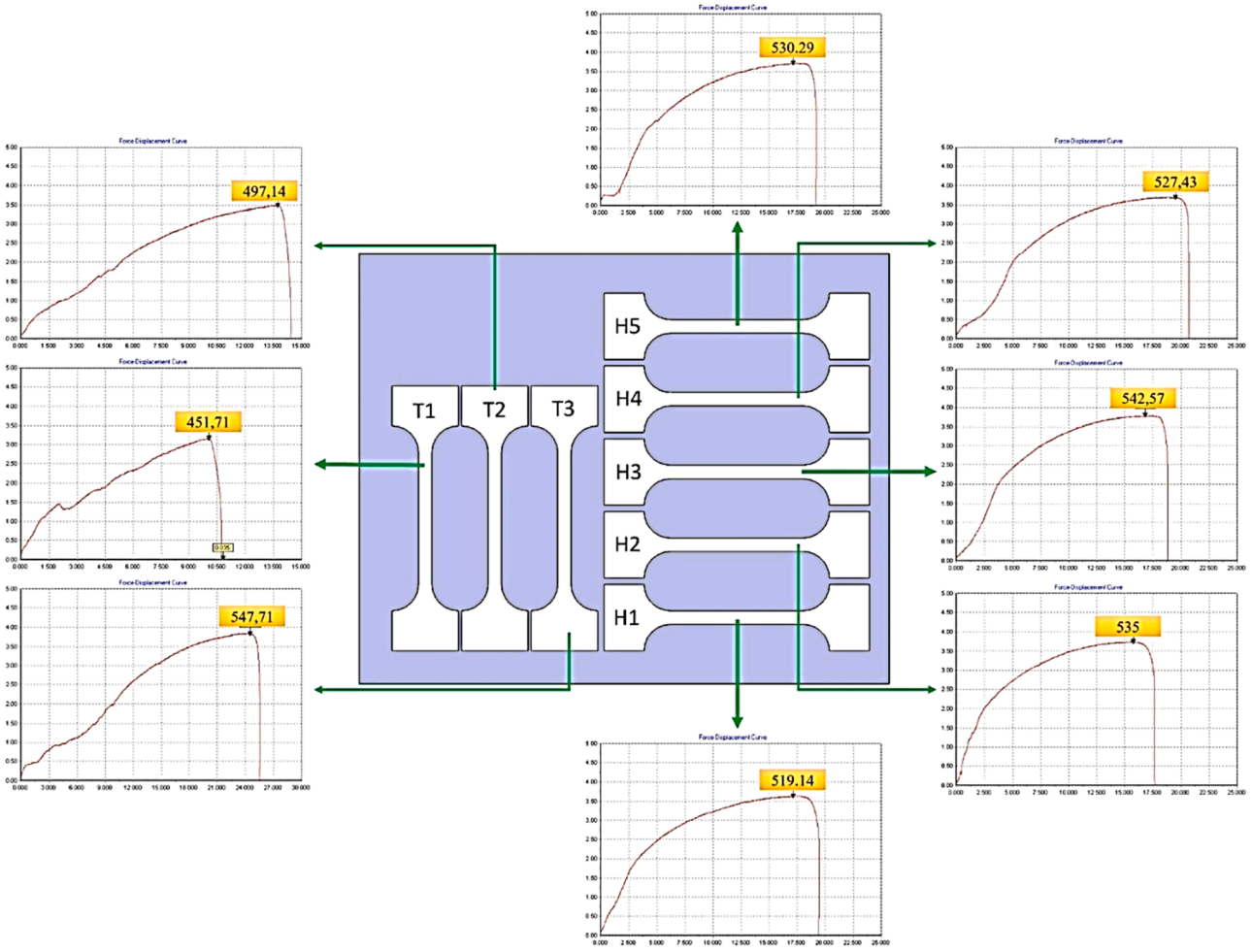


Fig. 7. Stress-strain diagram for each specimen.

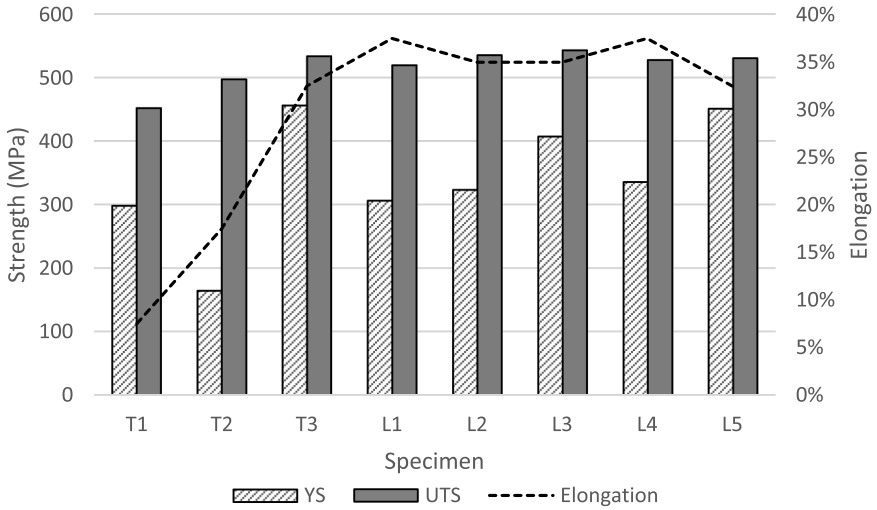


Fig. 8. The ultimate tensile strength (UTS), yield strength (YS), and elongation of specimens.

load-bearing applications.

3.5. Comparative analysis

The present TIG-WAAM investigation of 308 L stainless steel offers a comprehensive evaluation that encompasses microstructural evolution, mechanical properties, and importantly, corrosion resistance. This multifaceted approach distinguishes the current work from previous studies that focus solely on structural and mechanical aspects, while overlooking corrosion behavior, which is a critical property for stainless steel in service environments.

In the study conducted by Le and Mai (2020), 308 L stainless steel

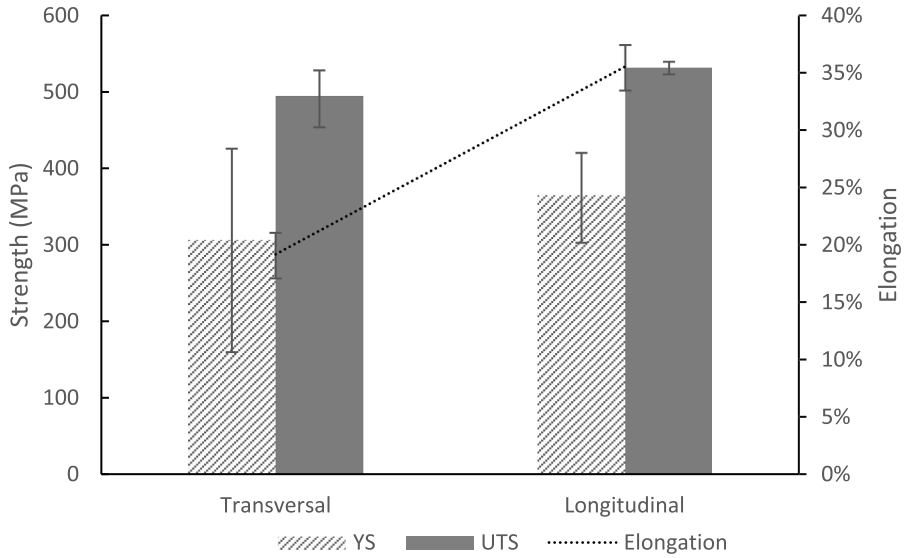


Fig. 9. Average of the ultimate tensile strength (UTS), yield strength (YS), and elongation.

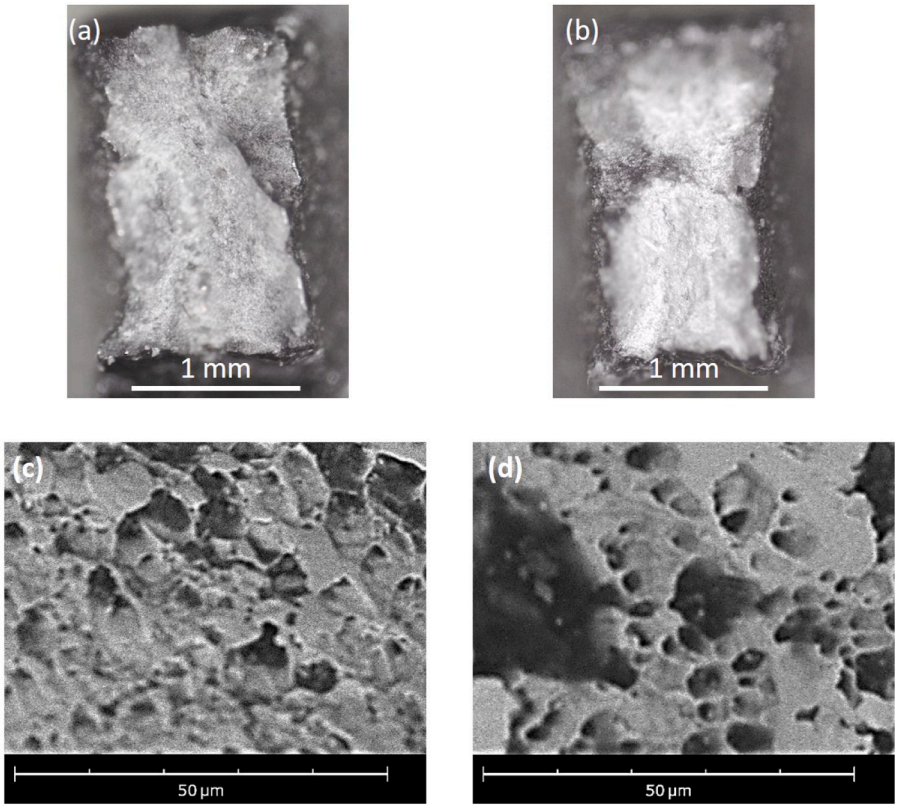


Fig. 10. Surface fracture, (a, c) transverse specimen (b, d) longitudinal specimen.

was deposited using Gas Metal Arc Welding-based Additive Manufacturing (GMAW-AM). Their microstructural analysis revealed predominantly columnar γ-austenite dendrites, interspersed with small amounts of δ-ferrite. The mechanical properties reported were in line with those of conventionally wrought 308 L, with tensile strength values of approximately 531.8 MPa (UTS) and 343.7 MPa (YS) in the vertical direction, and slightly higher values of 553 MPa (UTS) and 352.7 MPa (YS) in the horizontal direction. They also observed mechanical anisotropy, with the horizontal direction exhibiting greater ductility [40].

Similarly, Li et al. (2020) investigated the microstructure and mechanical behavior of 308 L stainless steel produced by Laminar Plasma Additive Manufacturing (LP-AM). Their study documented a progressive transformation in dendritic morphology along the build height, transitioning from columnar grains at the bottom to more equiaxed grains near the top due to changing thermal gradients. The LP-AM samples showed high build densities (~98.3 %) and improved isotropy in mechanical behavior at the top region [41].

In contrast, the TIG-WAAM results reported in this work fall within a comparable mechanical performance window, with UTS ranging from

518 to 547 MPa and YS from 342 to 364 MPa, depending on build orientation. As seen in the aforementioned studies, the present work also observed anisotropic mechanical behavior, with longitudinal (build-parallel) specimens generally outperforming transverse specimens in both strength and ductility. Microstructurally, the samples exhibited significant grain coarsening from the bottom to the top of the build, which is attributed to the cumulative heat input and reduced cooling rates at higher deposition layers. This trend is consistent with the grain morphology transitions observed in both GMAW-AM and LP-AM processes. However, the TIG-WAAM method showed more pronounced thermal accumulation effects, such as collapse zones and geometric distortion, suggesting a higher sensitivity to heat input control. This underscores the need for optimized thermal management and interpass cooling strategies in TIG-based AM processes to maintain build integrity and dimensional stability.

The most substantial contribution of this study lies in its inclusion of corrosion resistance assessment, a parameter absents in both previous works. Using Tafel polarization testing in a 3.5 wt. % NaCl solution, the corrosion rates were found to vary significantly along the build height. The bottom region, with finer grains and lower δ-ferrite content, exhibited superior corrosion resistance (0.0048 mm/year), while the top region showed significantly poorer performance (0.1112 mm/year), correlating with coarser grains and higher δ-ferrite fractions. These findings confirm the strong relationship between microstructural heterogeneity and electrochemical behavior, providing critical insight into the degradation mechanisms in arc-based additively manufactured stainless steel.

The absence of corrosion analysis in both Le & Mai (2020) [40] and Li et al. (2020) [41] represents a notable gap in the literature. Given that corrosion resistance is a defining property of stainless steels and essential for ensuring long-term durability in marine, chemical, and structural applications, its omission limits the practical applicability of their results. The present study not only bridges this knowledge gap but also emphasizes the necessity of integrating corrosion analysis into future research on stainless steel additive manufacturing.

The comparative analysis of the present TIG-WAAM results with those of previous GMAW-AM and LP-AM studies is summarized in Table 3. Explicitly, three distinctive aspects can be highlighted: (i) TIG-WAAM exhibits more pronounced interlayer heat accumulation than either GMAW-AM or LP-AM, which often results in collapse zones and geometric distortions. This observation indicates a higher sensitivity to heat input control, thereby emphasizing the need for optimized thermal management strategies in TIG-based processes, (ii) While grain coarsening along the build height is a common trend in arc-based AM, TIG-WAAM reveals a more significant microstructural gradient, ranging from fine dendrites at the bottom to coarse dendrites at the top. This pronounced gradient has a direct influence on both mechanical anisotropy and corrosion performance, and (iii) Unlike the previous GMAW-AM and LP-AM studies, the present work incorporates corrosion assessment, providing critical data that directly links microstructural heterogeneity with electrochemical behavior.

Table 3
Comparative Analysis of TIG-WAAM, GMAW-WAAM, and LP-WAAM Processes.

Aspect	This Study (TIG-WAAM)	Le & Mai (GMAW-AM)	Li et al. (LP-AM)
Power Source	GTAW (TIG)	GMAW	Laminar Plasma
Microstructure	Bottom: fine dendrites; top: coarse dendrites (δ-ferrite + γ)	Columnar dendrites + minor δ-ferrite	Dendritic austenite; grain size moderate/fine
Hardness (HV)	Bottom ~242; middle ~180; top ~170	155–169	Comparable to wrought steel
UTS / YS	Avg UTS ~519 MPa, YS ~342 MPa, max UTS ~547 MPa	Vertical UTS ~532 MPa & YS ~344 MPa; horizontal up to UTS ~553 MPa, YS ~353 MPa	Comparable values to wrought steel; orientation not specified
Anisotropy	Longitudinal (YS: 364 MPa, UTS: 531 MPa) > Transverse (YS: 306 MPa, UTS: 499 MPa)	Strong anisotropy: horizontal > vertical	Anisotropy not discussed
Grain Size Variation	Significant gradient bottom→top due to cooling rates	Columnar structure along build direction	Moderate grain variation, controlled structure
Unique Challenges	Interlayer heat accumulation → collapse zones & distortion	Thermal control critical for bead geometry	Thermal gradients moderated via plasma energy source
Corrosion behavior	Corrosion rate: top region > bottom region	Corrosion behavior not discussed	Corrosion behavior not discussed

4. Conclusion

This study evaluated the microstructural, mechanical, and corrosion properties of 308 L stainless steel fabricated using the Tungsten Inert Gas Wire Arc Additive Manufacturing (TIG-WAAM) process. The key findings can be summarized as follows:

- **Microstructural gradient and hardness variation:** Faster cooling at the bottom produced finer grains and higher hardness, while slower cooling at the top led to coarser grains and reduced hardness, underscoring the influence of thermal accumulation during deposition.
- **Anisotropic tensile behavior:** Mechanical testing revealed orientation-dependent properties, with longitudinal specimens demonstrating higher strength and ductility than transverse specimens, confirming the anisotropy inherent to the TIG-WAAM process.
- **Corrosion resistance gradient:** Electrochemical testing in 3.5 % NaCl solution showed superior corrosion resistance at the bottom (0.0048 mm/year) compared to the top region (0.1112 mm/year), directly linked to microstructural variation and δ-ferrite content.
- **Practical implications:** The results highlight the need for thermal management strategies, consideration of build orientation, and integration of corrosion performance evaluation when applying TIG-WAAM 308 L for critical industrial applications.

Limitations

This study did not include post-processing treatments, was based on a limited sample size, and did not assess long-term corrosion behavior. These aspects should be addressed in future investigations to enhance the reliability and applicability of TIG-WAAM 308 L components.

CRediT authorship contribution statement

Rauuf Nur Fattah: Writing – original draft, Investigation. **Iwan Istanto:** Formal analysis. **Bambang Kusharjanta:** Methodology. **Hammar Ilham Akbar:** Visualization. **Yupiter H.P. Manurung:** Conceptualization. **Moch Chamim:** Software. **Triyono:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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